

CHAPTER-5

Overview of Data Communication Networking

⇒ Types of Network

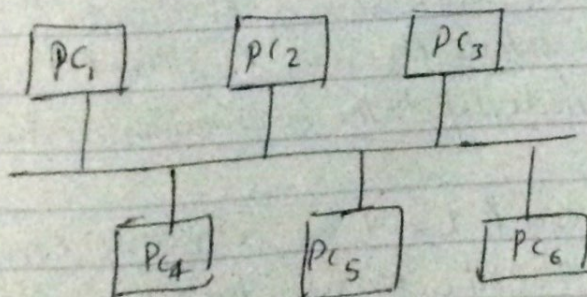
According to the distance covered by the NW, its physical architecture, its size & ownership, the n/ws are categorised in the following ways.

Network

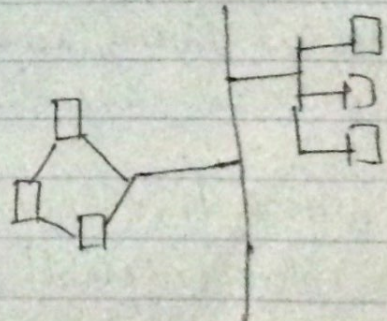
- LAN
- MAN
- WAN

(i) Local Area NW (LAN)

- multiple users computers connected together
- usually privately owned & links the devices in a single office building or campus.
- limited to few kilometers i.e. machines are spread over a small geographic region.
- Communication channels bet^h the machines are usually privately owned.
- channels are relatively error-free



LAN



② Metropolitan Area N/W (MAN)

- It spans over an entire city
- may be a single n/w such as cable TV N/W or a combination of different LANs into a large N/W to share the resources eg. banking N/W.
- May be wholly owned & operated by a private company or operated by public company eg. telephone company.

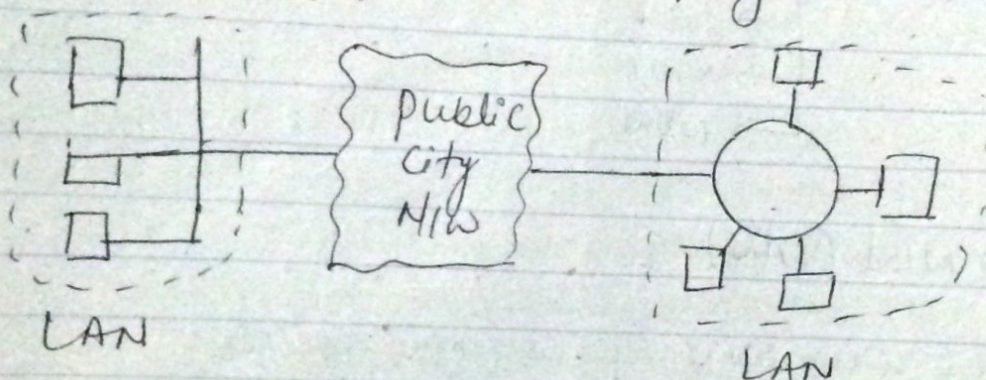


Fig. MAN.

③ Wide Area N/W. (WAN).

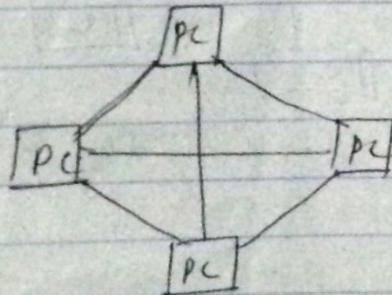
- Usually the wide area N/Ws cover a large geographical area. Typically a WAN consists of a no. of interconnected switching nodes.
- The largest WAN is the Internet, a collection of Networks & gateways linking millions of computers on every continent.
- A data is transmitted node to node up to the destination device.
- A WAN that is wholly owned & used by a single company is often referred to as an enterprise's n/w eg. IBM. owned by the public sector is called as internet.

Topology: -

- A topology of a n/w is the geometric representation of the relationship of all the links & linking devices
- There are five basic LAN Topologies.
mesh, star, tree, bus, & Ring.

(i) Mesh Topology

- In Mesh topology each device has a dedicated link to every other devices in the link
- Each link carries traffic only betⁿ the two devices it is connected by that link.



- In this topology $n(n-1)/2$ cable required to connect 'n' devices ie in above figure 6 cables required to connect 4 devices.

Advantage

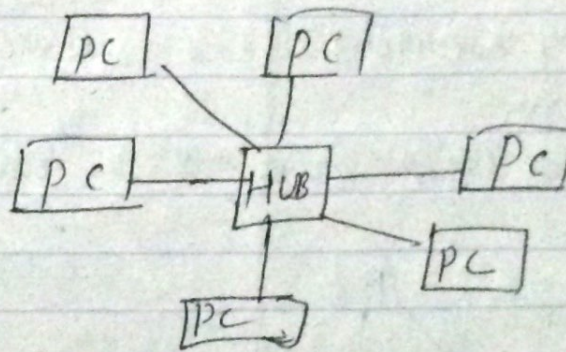
- Each connection can carry its own data load thus eliminate the traffic problem.
- Fault identification & fault isolation becomes easy.
- due to point to point connection privacy & security.

Disadvantage

- Large amount of cables are required.
- " no. of I/O ports are required.
- Installation & reconfiguration is difficult.

(11) Star Topology

- Each device has a dedicated point to point link only to a central controller usually called a hub.
- The devices are not directly linked to each other, the central device hub acts as an exchanger.
- If one device wants to send data to another device, it sends the data to the controller which then relays the data to other connected device.



(fig:- Star topology).

Advantage:

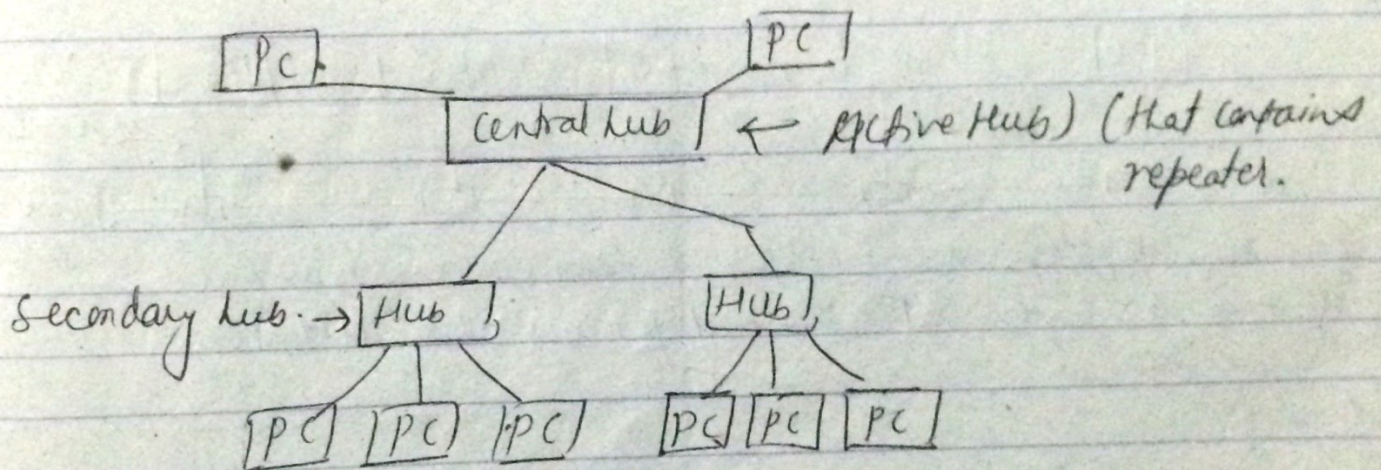
- less expensive than mesh topology.
- Easy installation.
- Easy fault identification & isolation.

Disadvantage

- Extra hardware is required.
- No. of devices of a network is limited by the no. of ports on the hub.
- Each device needs to be connected to a central hub.

II) Tree Topology (Extended star)

- In this topology nodes are linked to a central hub that controls the traffic to the n/w.
- It is modified star topology however not every device plugs directly into the central hub.



(fig. - Tree topology.)

Advantage :-

It allows more devices to be attached to a single central hub & also increases the distance that a signal can travel betⁿ devices.

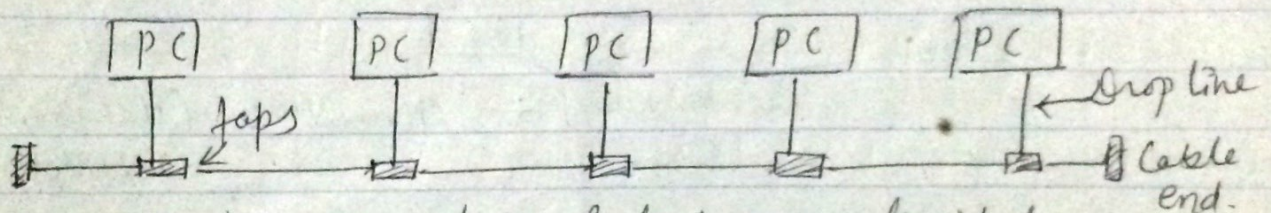
Disadvantage

- same as star topology

4 8 9 11 13

(iv) Bus Topology.

- It is multipoint topology.
- One long cable acts as a backbone to link all the devices in a n/w.
- nodes are connected to the bus cable by drop lines & taps.



- In Bus topology no. of taps & devices are limited & the distance betⁿ these taps is also limited.

advantage

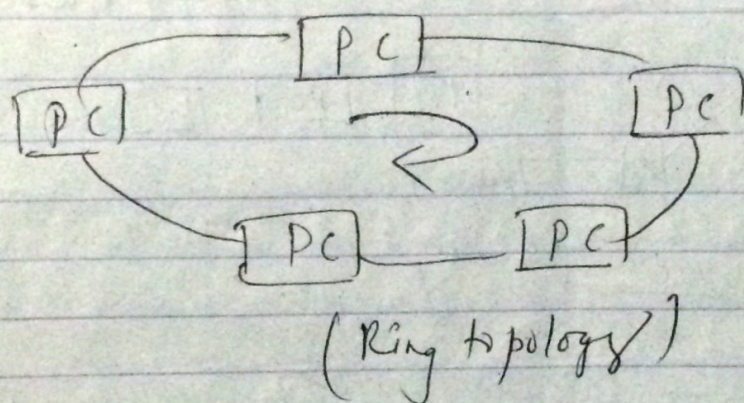
- Easy to install
- Less amount of cable
- cheaper

disadvantage

- Difficult in reconfiguration & fault isolation.
- " to add new devices.
- signal degradation at each tap.
- break on the backbone cable stops all transmission.

(v) Ring Topology

- Each device has a dedicated point to point connection only with two devices on either side of it.
- signal is passed along in one direction from device to device until it reaches its destination.
- Each device in a ring acts as a repeater



Advantage:

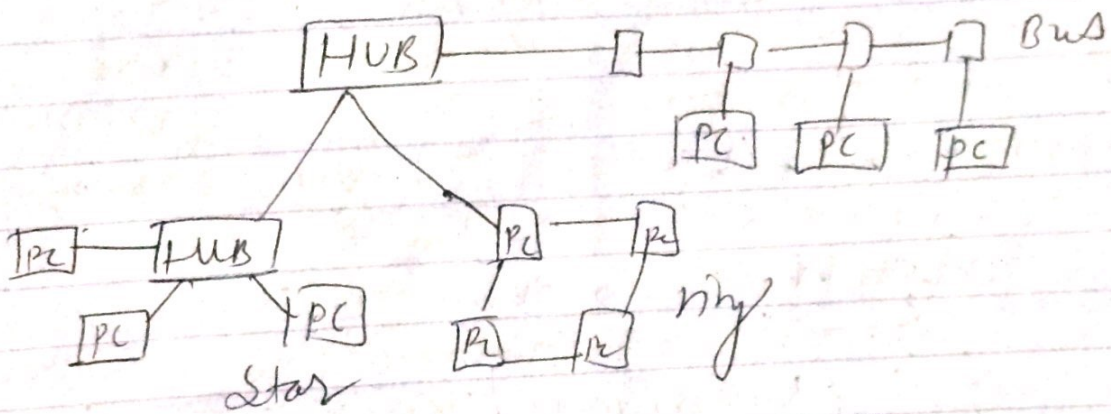
- Easy to install & re-configure
- fault isolation easy

Disadvantage

- unidirectional traffic
- A break in a ring can disable the entire n/w
- no. of device limited by ring length

Hybrid topology

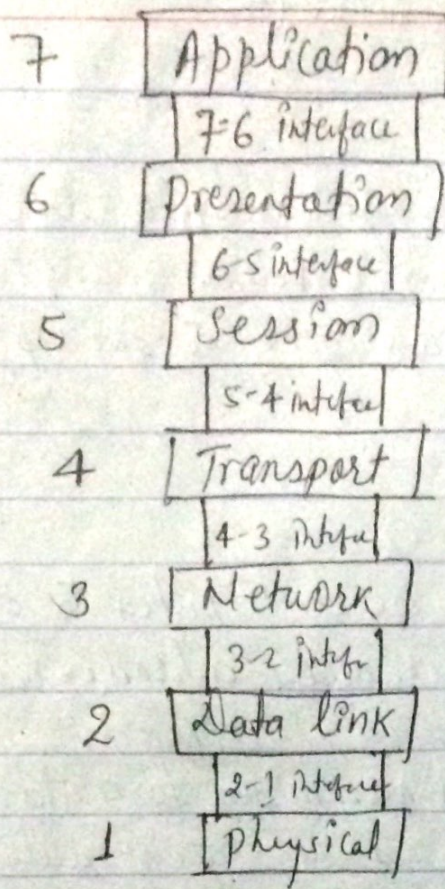
- In this topology, the n/w combines several topologies as sub-n/w, linked together in a large topology
- Different topologies can be connected to each other via a central controller in a star topology.



[fig:- Hybrid topology].

OSI layers & Functions. (Open System Interconnection)

- To place the distinct function in distinct place OSI is divided into layers so that the interconnection of these layers becomes a whole system, and the function of each layer is distinct so that no overhead to other mother layer from one layer.
- OSI model consists of seven separate but related layers.



(OSI Model)

easy way to remember the layers of OSI

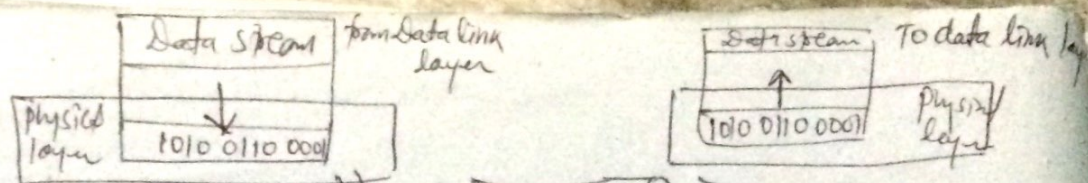
All People Seem To Need Data Processing.

आपों पेरी सूचना तिम्रो खकलि दशलि पहिने

A P S T M D P

① layer-1: physical layer.

- The physical layer co-ordinates the functions required to transmit a bit stream over a physical medium.
- Deals with the mechanical & electrical synchronization of the interface & transmission media. i.e. cable, plugs, pins, modulation, signal strength, voltage level.
- The main responsibility of this layer is to are
 - Bit signal representation
 - Bit synchronization
 - multiplexing
 - switching (circuit switching only).



② Layer-2 Data Link layer :-

- The data link layer transforms the physical layer to a reliable link and is responsible for node to node delivery.
- It makes the physical layer appear error free to the upper layer (ie N/w layer)
- Duties of data link layer

① Physical addressing :- If frames are to be distributed to different systems on the N/w the data link layer adds a header to the frame to define sender & receiver of the frame.

② Flow control :- It imposes the flow control mechanism to avoid overwhelming the receiver ie it maintains the rate of data sent by sender & received by receiver.

③ Error control :- This layer has mechanism to detect & retransmit damaged or lost frames

④ Access control :- When two or more devices are connected to the same link data link layer protocols are necessary to determine which device has control over the link at any given time.

③ Layer-3 Network layer

- This layer is responsible for the source-to-destination delivery of a packet across multiple N/w
- main duties

① Logical addressing :- It adds the header to the packet coming from the upper layer that include the logical address of the sender & receiver.

② Routing :- It choose the best path from different alternatives.

③ It helps to control congestion of packet

④ Transport layer - layer 4

- The transport layer is responsible for source to destination delivery of entire message from one process to another.
- A process is an application program running on a host

Duties

- ① The transport layer adds port address to the data so that the message goes to correct receiving computers.
- ② It segments & reassemble the message into transmittable segments. so that each segment contains a sequence no. which help to reassemble the segment at receiver side.
- ③ Also helps error & flow control.

⑤ layer - 5 Session layer

- The session layer is n/w dialog controller, it controls timing
- It establishes, maintains & synchronizes the interaction betⁿ communicating system
- It defines how the data conversations are established, controlled & terminated.

Main duties

- ① Dialog control: - It allows the communication betⁿ two process to take place either in half duplex or full duplex mode.
- ② synchronization: - Session layer allows a process to add checkpoints into stream of data due to which data is error free.

⑥ Layer - 6 Presentation layer:

- This layer is connected with the syntax & semantics of the information exchanged betⁿ two systems.

Syntax - Format or structure of data
Semantic - Meaning of data.

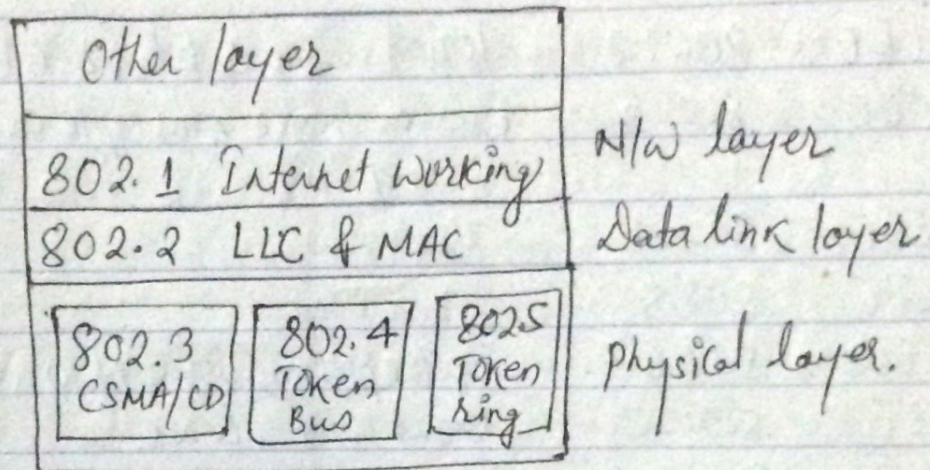
- Duties

- ① Translation :- This layer at the sender changes the infⁿ from its sender dependent format into a common format & viceversa.
- ② Encryption :- This layer encrypts the data at the sender & decrypts them at the receiver.
- ③ This layer compress the data to reduce the no. of bits to be transmitted.

⑦ Layer - 7 Application layer: (connecting betⁿ user & Internet)

- It provides user interfaces & supports for services such as e-mail, remote file access & transfer, share, database management & other distributed infⁿ services.
- This application layer provides the basis for email forwarding and storage.

LAN Architecture



(fig:- LAN Architecture).

LLC/MAC

- IEEE 802 standard divide OSI data link layer into two layer ie LLC & MAC

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(5W) * LLC (Logical link control) :-

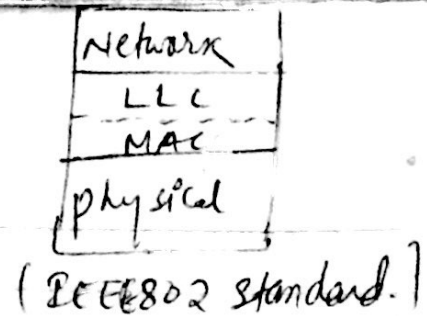
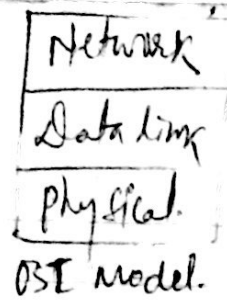
It is high level interface, which is associated with logical addressing between two users. and it specifies the mechanical form

* MAC (Medium Access control) / (Media access control)

- It is low level interface, which is associated with physical addressing & access control. eg - CSMA/CD.
- All LANs consists of devices that must share the N/w transmission capacity. MAC provide some means of controlling access to the transmission medium for an orderly & efficient use of that capacity.

IEEE standards Used in LAN architecture

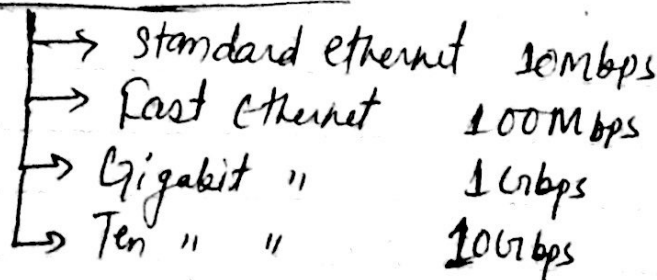
- | | | | |
|----|------|--------|---|
| 1) | IEEE | 802.2 | Logical Link Control (LLC) |
| 2) | IEEE | 802.3 | Carrier Sense Multiple Access / Collision Detection (CSMA/CD) |
| 3) | IEEE | 802.4 | Token Bus |
| 4) | IEEE | 802.5 | Token Ring |
| 5) | IEEE | 802.6 | Metropolitan Area N/W. |
| 6) | IEEE | 802.7 | Broadband LANs. |
| 7) | IEEE | 802.8 | Fiber optic LANs. |
| 8) | IEEE | 802.10 | Integrated data & voice N/W. |
| 9) | IEEE | 802.11 | Wireless N/W. |



Ethernet (CSMA/CD)

- Ethernet is standardized as IEEE 802.3, the combination of the twisted pair version for ethernet and connecting end system to the N/W, along with the Fiber optic versions for site backbone. and it is the most widespread wired LAN technology.
- Ethernet uses the CSMA/CD access method.

Ethernet evolution



2005

Q. How media is access using CSMA/CD.

2013

2012

CSMA/CD (Carrier sense multiple Access/collision Detection)

2010 (5)

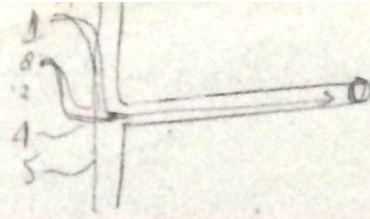
- It is the protocol used in ethernet n/w to ensure that only one network node is transmitting on the n/w wire at any one time.

- It is CSMA technique i.e. listening & waiting to transmit at a time & collision detection accidentally transmit at the same time, they are able to detect this error.

- If two stations sense the channel to be free & began transmitting simultaneously they will both detect the collision almost immediately without finishing transmission of frames. Then they should stop transmitting as soon as collision is detected. This protocol is known as CSMA/CD. & is widely used in LANs.

- For transmitting data through CSMA/CD protocol this leads following rule of transmission.

- (i) If the medium is idle, it transmit, otherwise go to step (ii)
- (ii) If the Medium (channel) is busy, continue to listen until the channel is idle then transmit immediately.
- (iii) If collision is detected during transmission, transmit a brief jamming signal to assure all stations to know that there has been collision occur & again transmit signal.



Routing :-

- Routing is the process of selecting paths in a network along which to send N/W traffic.
- Routing is performed for many kinds of N/W, including telephone N/W i.e. (circuit switching), electronic data N/W, Internet (packet switching)
- In packet switching N/W, routing directs packet forwarding towards their ultimate destination through intermediate nodes, typically N/W devices called routers, bridge, gateway, switches.
- The routing process usually direct forwarding packets on the basis of routing tables, which maintain a record of the routes to various N/W destination. Thus constructing routing table, which is held in routers memory & it is important for efficient routing.
- Most routing algorithms use only one N/W path at a time.
- There are various types of routing protocols defined for routing process eg static, Dynamic (RIP, EIGRP, OSPF etc)

Wide Area N/W

²⁰⁰⁹
X.25 :- It is a N/W protocol, developed in 1976 & it specifies an interface betⁿ a host system & packet switching N/W.

- It allows computers on different public networks to communicate through an intermediary computer at the N/W layer level

X.25 describes the procedure which are related for establishing, maintaining & terminating connections

The functionality of X.25 is specified on 3 levels

- 1) Physical layer level layer.
- 2) Link level.
- 3) packet level.

3	N/w layer	The packet level layer
2	Datalink layer	Link level layer
1	physical	physical level layer

Feature of X.25

(fig:- X.25 relation to the OSI Model)

- Both layer two & three include flow control & error control mechanism
- physical layer is called X.21 interface. The interface exist betw DTE & X.25 n/w
- X.25 provides data link control using a link access protocol balanced (LAPB) and transmits data as a sequence of frames.
- X.25 provides virtual circuit service. this service enables any subscriber to N/w to setup logical connection to other subscriber

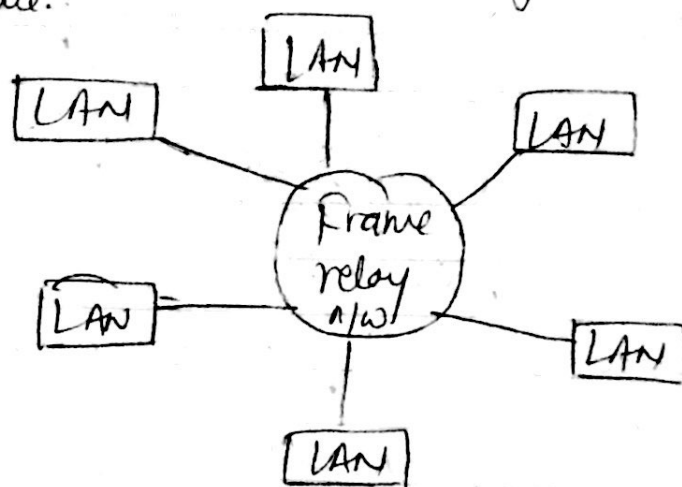
Drawbacks :-

- X.25 has a low 64kbps data rate
- It has extensive flow & error control at both data link layer & n/w layer which create a large overhead & slow down transmission.

2011F, 2010SP 2009SP

Frame Relay:-

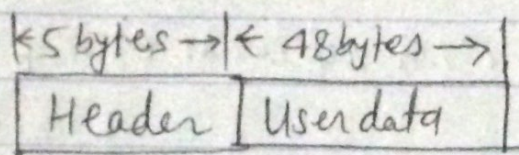
- Similar to X.25 but operates at a higher speed i.e. 1.544 Mbps & recently 44.376 Mbps.)
- Frame relay operates in just the physical & data link layers
- It is a telecommunication service designed for cost efficient data transfer for traffic betⁿ LANs & betⁿ WANs.
- It provides data in variable size unit called frame
- Less expensive than other traditional WANs.
- If the bad frame is received, the frame relay service discards that frame but in case of X.25 the acknowledgement is used for any damaged frame.



(fig:- using frame relay)

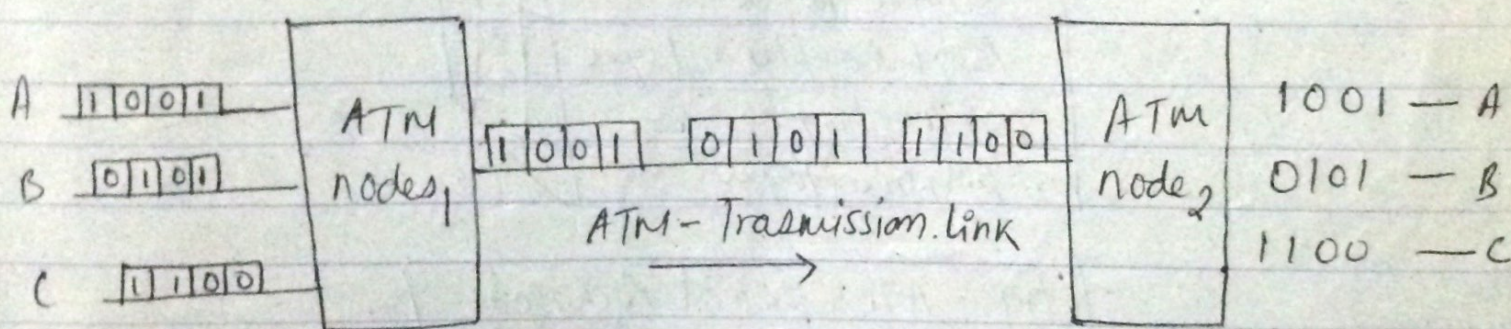
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ATM (Asynchronous Transfer Mode or module):



(fig:- ATM cell)

Sending
terminal

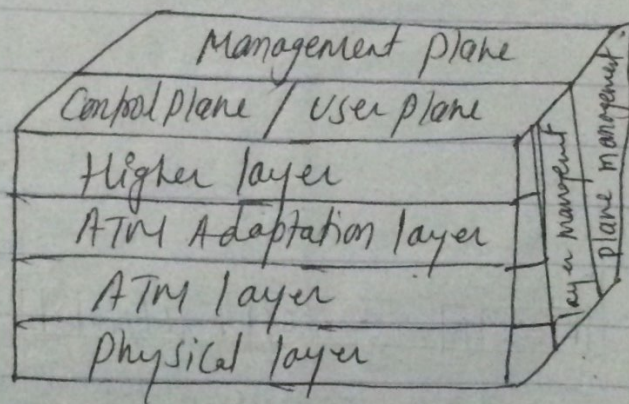


(fig:- data transmission mode of ATM)

- ATM is a frame relay protocol adopted by ITU-T
- ATM Sometimes refer to as cell relay.
- It can be viewed as an evolution from frame relay.
- The most obvious difference bet^h frame relay & ATM is that frame relay uses variable length of packets called frame, where ATM uses fixed length of packet called cell. Each cell is of 53 bytes long. Where 5 bytes for header & 48 bytes for data.
- ATM involves the transfer of data in discrete chunks at packet switching & frame relay.

- The data rate range from 25.6 Mbps to 622.08 Mbps.
- ATM is a streamlined protocol with minimum error & flow control capabilities

Protocol Architecture



(fig:- ATM protocol Architecture)